EXECUTIVE ANALYSIS

RETAIL ANALYTICS · PIPELINE REVIEW

Customer Shopping Behavior Analysis

From 3,900 raw transaction records to reproducible business actions using Python, SQL, and Power BI.

DATASET	3,900 customer purchases
FEATURES	18 behavioral & demographic fields
STACK	Python · PostgreSQL · Power BI

END-TO-END ANALYTICS WORKFLOW



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From transactional data to business decisions

SCOPE: 3,900 PURCHASES
STACK: PYTHON · SQL · POWER BI

01 / SCOPE

Analyzing retail transactions across four dimensions

- **Spending patterns:** transaction values and category averages.
- **Customer segments:** purchase frequency and customer loyalty.
- **Product preferences:** ratings, categories, and items.
- **Subscription behavior:** active member versus non-member trends.

FOUNDATION: 3,900 PURCHASES ACROSS PRODUCT LINES

02 / WORKFLOW

Three-stage analytical pipeline

STAGE 1 · INGESTION

Python & Pandas

- Profile missing values
- Impute category medians
- Feature binning & cleansing

DATA CLEANSING

CORE QUERIES

STAGE 2 · EXPLORATION

PostgreSQL Engine

- Revenue aggregations
- Frequency segmentation
- Discount & rating correlation

BUSINESS MODELING

STAGE 3 · DELIVERY

Power BI Dashboards

- Executive KPI tracking
- Interactive slicing
- Decision enablement

STAKEHOLDER INSIGHTS

DATA FLOW: RAW CSV → CLEANED DATABASE TABLE → INTERACTIVE VISUALIZATIONS

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Start with 3,900 rows and 18 columns

INPUT: RAW CUSTOMER DATA
QUALITY AUDIT.
CLEAN

01 / DATA DICTIONARY SNAPSHOT

Demographics	4 columns	Customer ID, Age, Gender, Location
Purchase Details	6 columns	Item Purchased, Category, Purchase Amount (USD), Size, Color, Season
Shopping Behavior	5 columns	Review Rating, Subscription Status, Shipping Type, Previous Purchases, Frequency of Purchases
Promotions	3 columns	Discount Applied, Promo Code Used, Payment Method

TOTAL SCHEMA: 18 COLUMNS COVERING BEHAVIORAL, TRANSACTIONAL, AND MARKETING ATTRIBUTES

02 / QUALITY AUDIT

IDENTIFIED GAP

37

Missing review ratings out of 3,900 entries (less than 1% null rate).

ACTION: IMPUTE CATEGORY MEDIAN IN PYTHON

03 / INTEGRITY CHECK

- No duplicate customer identifiers.
- Clean numerical ranges for age and spend.
- Consistent category nomenclature across seasons.

AUDIT RESULT: READY FOR FEATURE ENGINEERING

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Prepare the data in Python

STEP 01 / PROFILING

Inspect structure & distributions

Load CSV into pandas DataFrame; examine data types, nulls, and summary statistics to confirm data shapes.

IMPUTATION LOGIC

STEP 02 / CLEANING

Impute ratings & standardize headers

Fill the 37 missing ratings using category-specific medians; format column headers into clean snake_case.

STEP 03 / FEATURE ENGINEERING

Derive analytical dimensions

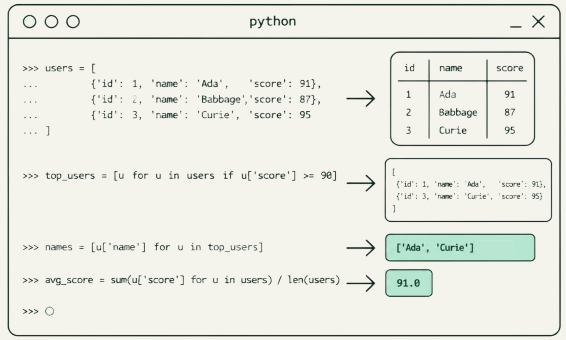
Create age groups and purchase-frequency days; check redundant discount and promo fields.

STEP 04 / DATABASE LOAD

Persist to PostgreSQL

Write transformed schema directly into production SQL tables with strict typing for analytical queries.

ENGINEERING ASSET



Clean & reproducible pipeline

Automated scripts guarantee consistent imputations, reproducible groupings, and validation checks before SQL execution.

SCHEMA: CUSTOMER_SHOPPING_DATA

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Use SQL to answer business questions

QUERY DOMAIN 01

Revenue across demographic cohorts

Compare total revenue and basket size by gender and age group to discover where customer purchasing power concentrates.

```
SELECT AGE_GROUP, GENDER, SUM(SPEND) ...
```

HIGH-VALUE SEGMENT

QUERY DOMAIN 03

Catalog performance & customer loyalty

Rank top-selling merchandise, track review scores, and segment customers by historical purchase counts to pinpoint retention.

```
SELECT ITEM_PURCHASED, COUNT(*), AVG(RATING) ...
```

QUERY DOMAIN 02

Fulfillment & subscription contribution

Measure revenue splits across standard versus express shipping types and evaluate top-line revenue by subscription status.

```
SELECT SHIPPING_TYPE, SUBSCRIPTION_STATUS ...
```

QUERY DOMAIN 04

Discount penetration & order sizes

Quantify discounted order percentages and identify customers spending above average while utilizing promotional offers.

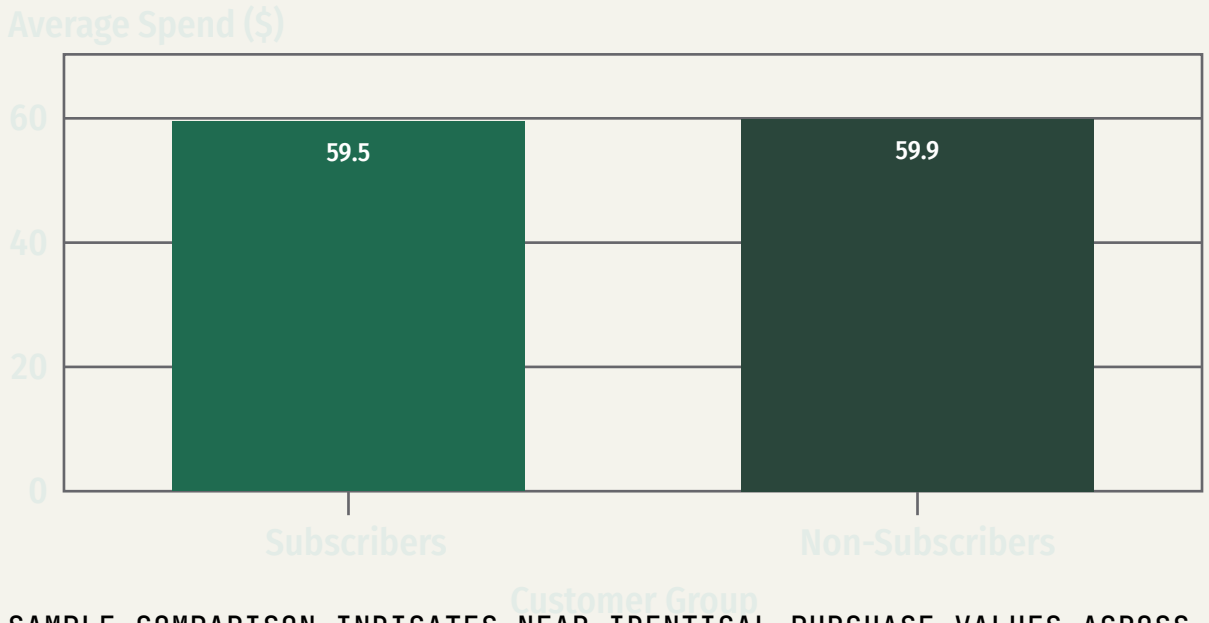
```
WHERE DISCOUNT_APPLIED = 'YES' AND SPEND > AVG_SPEND ...
```

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Subscribers and non-subscribers spend similarly

SAMPLE: 3,900 CUSTOMERS
FINDING: PARITY IN BASKET SIZE

01 / AVERAGE SPEND PER PURCHASE (USD)



SAMPLE COMPARISON INDICATES NEAR-IDENTICAL PURCHASE VALUES ACROSS COHORTS.

02 / REVENUE BREAKDOWN

KEY INSIGHT

Subscribers	1,053	$62,645
Non-Subscribers	2,847	$170,436

TOTAL SAMPLE REVENUE: $233,081

03 / COMMERCIAL NOTE

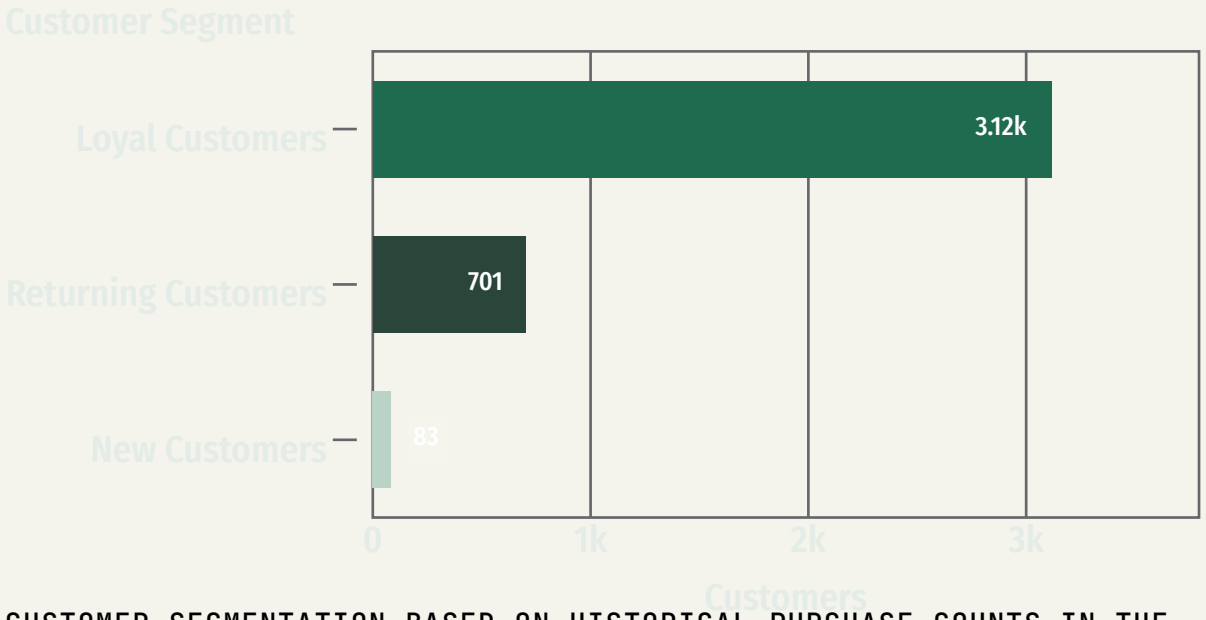
Average transaction size is virtually identical ($59.49 vs. $59.87). Non-subscribers generate 73% of sample revenue simply due to volume, showing subscription does not imply higher basket size.

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Purchase history supports customer segmentation

METRIC: PREVIOUS PURCHASES
LOYALTY SHARE
79.9%

01 / CUSTOMER DISTRIBUTION BY SEGMENT



CUSTOMER SEGMENTATION BASED ON HISTORICAL PURCHASE COUNTS IN THE DATASET.

02 / SUBSCRIPTION PENETRATION

REPEAT COHORT

High-frequency buyers

Among customers with more than five previous purchases:

3,476

total repeat shoppers in this group

- **2,518** are non-subscribers (72.4%)
- **958** are active subscribers (27.6%)

UNCONVERTED LOYALTY OPPORTUNITY

03 / TAKEAWAY

The vast majority of frequent, loyal customers have not enrolled in subscriptions, highlighting an organic retention base that can be converted with targeted incentives.

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RETAIL ANALYTICS · BOARD 08

SQL highlights product and discount patterns

DISCOUNT THRESHOLD: ABOVE AVG
TOP ITEMS: GLOVES & HATS

01 / HIGHEST CUSTOMER RATING

Gloves

Gloves earn the highest listed average review rating across product categories.

3.86

AVERAGE RATING

02 / HIGHEST DISCOUNT RATE

Hats

Hats record the highest listed discounted-purchase rate in the merchandise catalog.

50%

DISCOUNTED SHARE

03 / DISCOUNT USERS

839

Discount users spent above the average purchase amount ($59.76).

SQL QUERY: SPEND > OVERALL_AVG

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Power BI brings the findings together



OVERALL KPIS

SUMMARY CARDS

$59.76

AVG PURCHASE AMOUNT

3.75

AVG REVIEW RATING

DASHBOARD CAPABILITIES

- **Cross-filtering:** dynamic slicing by subscription status, age cohorts, and category.
- **Revenue tracking:** visual breakdown of sales contributions across product lines.
- **Self-serve analysis:** interactive filters for operational and category managers.

INTEGRATED BI REPORTING LAYER

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Turn findings into business actions

RETENTION

ACTION 01 / LOYALTY

Promote subscriber benefits

- Add tangible perks since basket size alone does not distinguish subscribers.
- Reward repeat buyers and convert the 2,518 un-enrolled frequent buyers with targeted loyalty rewards.

FOCUS: REPEAT SHOPPER CONVERSION

ACTION 02 / PRICING

Review discount policy

- Enforce margin controls on heavily discounted lines like Hats (50% promo rate).
- Structure promotional minimum spend tiers around the 839 high-basket discount users.

FOCUS: MARGIN PRESERVATION

ACTION 03 / MERCHANDISING

Merchandising & targeting

- Highlight top-rated and best-selling products, such as Gloves (3.86 rating), across storefront channels.
- Target high-revenue age brackets identified in SQL queries for seasonal campaigns.

FOCUS: PRODUCT PROMOTION

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